

☒ Parking Pass Tracker

A QR code-based check-out system for temporary parking passes: automated tracking, reminders, and overdue alerts for a leasing office.

HTML/CSS · JavaScript · Google Apps Script · Google Sheets · GitHub Pages · Process Automation

☒ The Problem

During my internship at a leasing office, one of the most frustrating recurring problems was the temporary parking passes.

- Each pass is custom-made to work with the property's parking system and costs \$50 to replace.
- The office kept about 10 passes in rotation, about \$500 in equipment.
- When the front desk handed out a pass, staff mostly had to **remember** who had it.
- Front desk employees work **different days**, so the knowledge didn't carry over from shift to shift.
- We tried logging passes in the resident CRM (**Entrata**), but our setup had **no automatic reminders or follow-ups**, so it still depended on someone remembering to check.
- Residents signed a **paper form**, but those forms were often **lost between workers**.

The result: passes went missing regularly, and the property was losing close to \$500 almost every month in replacement costs, along with staff time spent tracking passes down.

☒ The Solution

I built a simple, free, automated system that takes memory and paper out of the process:

1. **Scan.** A resident scans a QR code at the front desk.
2. **Request.** A 2-page web form collects their name, room number, phone, email, and borrow date.
3. **Agree.** Page 2 shows the **due date (14 days later)** and a liability agreement. The resident signs by typing their name, acknowledging that an unreturned pass may lead to fees including, but not limited to, a **\$200 parking fee and possible towing**.
4. **Log.** The request is saved automatically as a row in a shared spreadsheet that every staff member can see.
5. **Confirm.** The resident gets a confirmation email and management gets a "new pass issued" alert.

6. **Remind.** 24 hours before the due date, the resident gets a reminder (management CC'd), which says to disregard it if the pass is already returned.

7. **Escalate.** If the pass still isn't marked returned after the due date, management gets an "ACTION NEEDED" email to take action and review the resident's CRM profile.

When the pass comes back, staff set **Pass Returned?** → **Yes**, and all reminders stop.

⚙ What It Does

Feature	Description
QR code access	Residents open the form from their phone. No app needed
Digital agreement	Typed signature must match the name entered, so there's a clear record
Auto due date	Calculated as borrow date + 14 days
Central log	Every pass is in one shared sheet, so nothing depends on one person's memory
Unique Pass IDs	Each checkout gets an ID (e.g., PP-261001-5)
Color-coded status	🟢 Returned · 🟡 Due within 24 hrs · 🔴 Overdue
Automated emails	Confirmation, 24-hour reminder, and overdue escalation
Hourly check	Time-based trigger checks every pass automatically
Excel export	File → Download → .xlsx anytime

🔧 Tech Stack

- **Frontend:** HTML, CSS, JavaScript (hosted free on GitHub Pages)
- **Backend:** Google Apps Script web app
- **Database:** Google Sheets (data validation, conditional formatting)
- **Automation:** Apps Script time-driven triggers + MailApp email service
- **Cost to run:** \$0

🔧 How It Works

QR Code → HTML Form (GitHub Pages) → Apps Script Web App → Google Sheet



🔧 Files

File	Purpose
index.html	2-page resident form + agreement
Code.gs	Backend: saves data, sends emails, runs reminders

File	Purpose
README_SETUP.txt	Step-by-step deployment guide

☒ Setup

See [README_SETUP.txt](#). Setup takes about 20 minutes:

1. Create a Google Sheet → Extensions → Apps Script → paste `Code.gs` → run `setup()`.
2. Deploy as a Web App and copy the URL.
3. Paste the URL into `index.html` and publish it on GitHub Pages.
4. Generate a QR code for the Pages link and post it at the front desk.

☒ Impact

- Replaces lost paper forms and memory-based tracking with **one shared source of record**.
- Works across **every shift**, since any staff member can see who has each pass.
- Automatic reminders mean residents return passes **before** they become a problem.
- Built to prevent losses of up to **~\$500/month** in replacement passes.

☒ Future Improvements

- Migrate the data to a SQL database or Microsoft Azure for reporting.
- Rebuild with Microsoft Forms + Power Automate for Microsoft 365 offices.
- Add an SMS reminder option.
- Add a dashboard for pass usage trends and return rates.

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